



RB001 Week 6 — English Worksheet

Topic: Coins & the Leaderboard · **Course:** Obby Architect · **Time:** about 45 minutes

This week your obby gets **coins**. When a player joins, your script builds a **leaderstats** folder with a **Coins** value. When a player touches a coin, Coins goes up by 1 and the coin disappears.

1 · Predict 🧠

Read each script. Before you imagine running it in Studio, write what you think will happen.

```
local Players = game:GetService("Players")

Players.PlayerAdded:Connect(function(player)
    local leaderstats = Instance.new("Folder")
    leaderstats.Name = "leaderstats"
    leaderstats.Parent = player

    local coins = Instance.new("IntValue")
    coins.Name = "Coins"
    coins.Parent = leaderstats
end)
```

**You press Play and join the game. What shows on the leaderboard in the top-right corner?
What number does Coins start at?**


```

local Players = game:GetService("Players")
local coin = script.Parent

coin.Touched:Connect(function(otherPart)
    local player = Players:GetPlayerFromCharacter(otherPart.Parent)
    if player then
        print("a player touched the coin")
    else
        print("not a player")
    end
end)

```

A loose part rolls into the coin. Which message prints? Then you walk into the coin yourself. Which message prints now?


```

local Players = game:GetService("Players")
local coin = script.Parent
local collected = false

coin.Touched:Connect(function(otherPart)
    local player = Players:GetPlayerFromCharacter(otherPart.Parent)
    if player and not collected then
        collected = true
        player.leaderstats.Coins.Value += 1
        coin:Destroy()
    end
end)

```

You step on this coin. What happens to your Coins number? What happens to the coin itself?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the script is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — When a player joins, Coins should appear on the **leaderboard**. The script runs with no errors, but the leaderboard never shows up.

```
local Players = game:GetService("Players")

Players.PlayerAdded:Connect(function(player)
    local leaderstats = Instance.new("Folder")
    leaderstats.Name = "MyStats"
    leaderstats.Parent = player

    local coins = Instance.new("IntValue")
    coins.Name = "Coins"
    coins.Parent = leaderstats
end)
```

Hint: Roblox only shows the folder if its name is **exactly** right.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — One coin should give **exactly 1** point. But when you run through the coin, you sometimes get 2 or 3 points from one coin.

```

local Players = game:GetService("Players")
local coin = script.Parent

coin.Touched:Connect(function(otherPart)
    local player = Players:GetPlayerFromCharacter(otherPart.Parent)
    if player then
        player.leaderstats.Coins.Value += 1
        coin:Destroy()
    end
end)

```

Hint: `Touched` fires once for each body part. You need a `local collected = false` guard.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The coin should give a point only to a **player**. But when a loose part bumps the coin, the Output window shows a red error and the coin breaks.

```

local Players = game:GetService("Players")
local coin = script.Parent
local collected = false

coin.Touched:Connect(function(otherPart)
    local player = Players:GetPlayerFromCharacter(otherPart.Parent)
    collected = true
    player.leaderstats.Coins.Value += 1
    coin:Destroy()
end)

```

Hint: when the toucher is not a player, `player` is `nil`. Check before you use it.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your obby in Roblox Studio. Add a `leaderstats` script with Coins, then place at least 3 coins in your obby that each give 1 point and disappear. Collect them all in Play mode. When you finish, come back here.

Send a photo or video of your leaderboard showing your Coins, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The leaderboard works because the folder is named ...

I used `GetPlayerFromCharacter` to ...

My debounce **collected** stops ...

One tricky moment was when ...

To fix it, I ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you collect your coins in Play mode. Talk through it like you are teaching someone who has never seen it. Try to use these words: **leaderboard**, **player**, **value**, **debounce**, **destroy**.

1. Press Play and point at the leaderboard with Coins on it.
2. Collect one coin and show the number go up by exactly 1.
3. Read your coin script out loud and say what `if player and not collected` checks.
4. Say in your own words why the coin is destroyed at the end.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.